AYANA K K

AI ENGINEER | RESEARCH ASSOCIATE

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in **Large Language Models, Agentic AI,** and **Edge AI**. Proven ability to design, optimize, and deploy **intelligent systems** that automate complex workflows, enable accurate knowledge retrieval at scale, and perform efficiently on resource-constrained hardware. Effective communicator, able to **translate complex technical concepts** for both technical teams and business stakeholders. Committed to building efficient, scalable, production-ready AI solutions, that drives measurable business value.

SKILLS

- **Programming:** Python, SQL, C/C++
- **AI & LLMs:** Agentic AI, LLMs, RAG, LangChain, AI Workflows, Retrieval Evaluation
- **Machine Learning & Deep Learning:** Scikit-learn, XGBoost, Ensemble Learning, Feature Engineering, PyTorch, Keras, CNNs, Transfer Learning, Model Compression
- **Computer Vision & Edge AI:** Semantic Segmentation, Object Detection, Pruned CNNs, BNNs, AdderNet, DeepShift
- **Backend & AI Infrastructure:** FastAPI, REST APIs, Async Python (asyncio), React, FFmpeg, ElevenLabs, FAISS, Cross-Encoder Re-ranking
- **Data & Databases:** GitHub Actions CI, pytest, Structured Logging, Caching, Pandas, NumPy, Matplotlib, Power BI, MySQL

WORK EXPERIENCE

Research Associate *Feb 2026 – Present*
ReyraAI, Kozhikode

- Architected a **full-stack asynchronous AI video dubbing platform (FastAPI, React)** with a job-based REST API and async status polling, supporting multi-minute GenAI workloads without client timeouts. Implemented **concurrent multi-language processing** and fault-isolated pipelines, reducing batch turnaround, and automated **frame-accurate ffmpeg editing, AI voice-track muxing, and SRT/VTT subtitle generation**.
- Designed and deployed a **production-grade Retrieval-Augmented Generation (RAG) system** using LangChain, FAISS, and Ollama-hosted LLMs, containerized with Docker Compose and served via FastAPI. Optimized vector search (IVF/HNSW indexing) with **cross-encoder re-ranking** to cut query latency and lift Recall, backed by structured logging, caching, and CI/CD pipelines (pytest, GitHub Actions).
- Explored **Machine Unlearning techniques** to selectively remove the influence of specific training data from trained models without full retraining, supporting model compliance and privacy-preserving deployment.
- Engineered **energy-efficient deep learning architectures** for Edge AI, including Pruned CNNs, Binary Neural Net- works (BNNs), AdderNet, and DeepShift, applying **model compression** (structured pruning, binarization, shift-based quantization) to meaningfully shrink model footprint and accelerate inference with minimal accuracy loss.
- Built a **reproducible PyTorch benchmarking framework** evaluating accuracy, latency, FLOPs, memory, and energy efficiency across multiple model variants, driving hardware-aware architecture selection for low-power, on-device AI deployment.

Data Science & AI Trainer *Jun 2025 – Dec 2025*
Maitexa Technology, Kozhikode

- Conducted hands-on training sessions on Python, SQL, Data Analysis, Machine Learning, Deep Learning, and Power BI.
- Guided students in data preprocessing, SQL querying, feature engineering, data visualization, and Power BI dashboard development.
- Mentored students on machine learning workflows, model evaluation, practical projects, and data-driven problem solving.

PROJECTS

D Triage Severity & ICU-Risk Classification System | *Python, scikit-learn, FastAPI, React, Vite*

- Built a **5-class ESI triage ML pipeline**, removing **28+ leakage-prone features** and evaluating Logistic Regression, Random Forest, and Gradient Boosting with **stratified k-fold CV, weighted Cohen's Kappa,** and **error analysis**.
- Developed and deployed a **FastAPI + React/Vite clinical dashboard** with **permutation feature importance (XAI)**, hosted on **Render and Vercel** with secured CORS configuration.

AI Video Dubbing Platform | *FastAPI, React/TypeScript, FFmpeg, asyncio*

- Built a **full-stack asynchronous video dubbing platform** with **FastAPI and React/TypeScript**, using job-based REST APIs and async polling for non-blocking AI processing.
- Implemented **concurrent multi-language processing with asyncio.gather** and an **FFmpeg media pipeline** for frame-accurate trimming, AI voice-track muxing, and automated **SRT/VTT subtitle generation**.

Production RAG System | *LangChain, FAISS, Ollama, FastAPI,*

- Built and deployed a **production-style RAG system** using **LangChain, FAISS, and Ollama**, exposed through **FastAPI** and containerized with **Docker Compose** for reproducible deployment.
- Optimized retrieval with **FAISS IVF/HNSW indexing and cross-encoder re-ranking**, improving **Recall@k** and **query latency** based on a custom evaluation harness; added **query caching, structured logging, and automated testing**.

Voice-Driven Lead Generation System | *ElevenLabs, FastAPI, Google Sheets API*

- Built an end-to-end **voice-driven lead generation system** with multi-turn customer discovery, FastAPI business logic, **knowledge-grounded responses**, and webhook-based tool calling to convert conversations into structured lead records.
- Integrated **Pydantic validation and Google Sheets API** with OAuth2 for reliable lead persistence and real-time sales visibility, using a **modular, provider-agnostic architecture** for scalable integrations.

EDUCATION

Master of Science — Computer Science (Intelligent Systems)
Central University of Kerala | *Jul 2023 – Jun 2025*

Bachelor of Science — Computer Science
Mahatma Gandhi College, Iritty (Kannur University) | *May 2019 – May 2022*